



**INTELLIGENT FIELD SURVEILLANCE AND PRECISION
CONTROL FRAMEWORK WITH OPTIMIZED YOLO**

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Current scenario in agriculture is highly affected by pest infestations that significantly reduce crop yield and quality, leading to major economic losses for farmers worldwide. Traditional pest monitoring methods depend heavily on manual field inspection, which is slow, labor-intensive, and often inaccurate, especially when dealing with large farmlands. These methods are also ineffective in detecting small, overlapping, or visually similar pests in real-time environmental conditions, resulting in delayed response and increased crop damage. As a result, there is a strong need for an automated, intelligent, and real-time pest detection system that can assist farmers in making timely and accurate decisions. To address these challenges, an advanced real-time pest detection system is proposed using an enhanced YOLO-based deep learning framework integrated with attention mechanisms and residual connections for improved detection accuracy. The system is capable of identifying multiple pest types in complex agricultural environments and supports real-time deployment on drones, mobile devices, and field surveillance cameras. This approach enables early pest identification, minimizes chemical misuse, improves crop protection efficiency, and promotes sustainable and eco-friendly smart agriculture practices.

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LIST OF ABBREVIATIONS

AI	- Artificial Intelligence
AJAX	- Asynchronous JavaScript and XML
API	- Application Programming Interface
CNN	- Convolutional Neural Network
CPU	- Central Processing Unit
CSS	- Cascading Style Sheets
DOM	- Document Object Model
Flask	- Python Web Framework
GPU	- Graphics Processing Unit
GUI	- Graphical User Interface
HTML	- HyperText Markup Language
IDE	- Integrated Development Environment
IoT	- Internet of Things
JSON	- JavaScript Object Notation
KNN	- K-Nearest Neighbors
NPK	- Nitrogen, Phosphorus, Potassium
OpenCV	- Open Source Computer Vision Library

PE	- Professional Edition
PyCharm	- Python Integrated Development Environment
RAM	- Random Access Memory
RDBMS	- Relational Database Management System
SQL	- Structured Query Language
SSM	- State Space Model
SMTP	- Simple Mail Transfer Protocol
SVM	- Support Vector Machine
TensorFlow	- Open-Source Machine Learning Framework
URL	- Uniform Resource Locator
YOLO	- You Only Look Once
YOLOv8	- You Only Look Once Version 8

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Agriculture plays a crucial role in global food security and economic development, but it is constantly threatened by pest infestations that reduce both crop yield and quality. In the current scenario, farmers largely depend on manual inspection methods to identify pest attacks in fields. These traditional approaches are time-consuming, labor-intensive, and often unreliable when applied to large-scale agricultural areas. As pests can spread rapidly under favourable conditions, delayed detection often results in severe crop damage and financial loss, highlighting the need for faster and more accurate detection systems. In recent years, technological advancements in artificial intelligence and deep learning have opened new possibilities for transforming agricultural practices. Among these, object detection models such as YOLO (You Only Look Once) have gained significant attention due to their ability to perform fast and accurate real-time detection.

However, standard detection models may still struggle with challenges such as detecting small-sized pests, overlapping insects, and visually similar species in complex natural backgrounds. These limitations make it necessary to enhance model performance for practical agricultural deployment. To overcome these challenges, an improved YOLO-based deep learning framework is proposed by integrating attention mechanisms and residual connections. These enhancements improve the model's ability to focus on relevant pest regions while preserving important feature information, enabling more accurate detection and classification. Furthermore, the system extends beyond detection by incorporating an intelligent pesticide recommendation module, which suggests appropriate chemical solutions based on the identified pest type. This integrated

approach supports real-time monitoring, early intervention, and sustainable farming practices, ultimately contributing to smarter and more efficient agricultural management.

1.1.1 OVERVIEW

The proposed project focuses on developing an intelligent real-time pest detection and pesticide recommendation system for smart agriculture using an enhanced YOLO-based deep learning framework. In the current agricultural scenario, pest detection is largely dependent on manual observation, which is inefficient, time-consuming, and unreliable for large-scale farming. To overcome these limitations, the system is designed to automatically detect and classify different types of pests from images or live video streams captured using drones, mobile devices, or field surveillance cameras, ensuring continuous and accurate monitoring of crop fields. The core of the system is an improved YOLO (You Only Look Once) model integrated with attention mechanisms and residual connections to enhance detection performance.

These enhancements allow the model to effectively identify small-sized pests, overlapping insects, and visually similar species even in complex agricultural environments. The system processes real-time input data and performs fast object detection, enabling immediate identification of pest presence with high accuracy and reliability, which is essential for timely agricultural intervention. In addition to pest detection, the project incorporates an intelligent pesticide recommendation module that suggests suitable pesticides or fertilizers based on the identified pest species. This module is built using a knowledge-based mapping system that links detected pests with appropriate treatment solutions, including dosage and usage guidelines. By combining real-time detection with decision support, the system reduces dependency on agricultural experts, minimizes chemical misuse, and promotes sustainable and eco-friendly farming practices, ultimately improving crop health and productivity.

1.1.2 IMPORTANCE AND IMPACTS

The importance of this project lies in its ability to address one of the most critical challenges in agriculture effective and timely pest management. Pest infestations are a major cause of crop loss, and traditional detection methods are often slow and inefficient. By introducing a real-time, automated pest detection system using an enhanced YOLO-based deep learning framework, the project significantly improves the accuracy and speed of pest identification. This enables farmers to monitor large agricultural areas efficiently and detect infestations at an early stage, reducing the risk of severe crop damage and economic loss. The project also has a strong impact on decision-making in agriculture through its intelligent pesticide recommendation system. Farmers often lack access to immediate expert guidance, leading to incorrect pesticide usage and excessive chemical application.

By providing precise recommendations based on detected pest species, the system ensures that the right type and amount of pesticide is used. This not only improves pest control effectiveness but also reduces input costs for farmers. Additionally, minimizing chemical overuse helps protect soil health, water resources, and biodiversity, contributing to more sustainable agricultural practices. Furthermore, the project plays a vital role in promoting smart and precision agriculture by integrating advanced technologies such as deep learning, computer vision, and real-time monitoring systems. It supports modern farming techniques by enabling deployment on drones, mobile devices, and surveillance systems, making it scalable and adaptable to various farming conditions. The overall impact includes increased crop productivity, improved resource management, reduced environmental impact, and enhanced farmer livelihoods. This makes the system a valuable contribution toward achieving sustainable and technology-driven agricultural development.

1.1.3 BACKGROUNDS

Agriculture has always been a fundamental sector for sustaining human life and supporting economic development, but it faces continuous challenges due to pest infestations that negatively impact crop yield and quality. For decades, farmers have relied on traditional pest detection methods such as manual field inspection and visual observation. While these methods have been widely used, they are often inefficient, time-consuming, and prone to human error, especially in large-scale farming environments. As agricultural practices expand and intensify to meet growing food demands, the limitations of conventional pest monitoring techniques have become increasingly evident. With the advancement of technology, efforts have been made to improve pest detection through basic image processing and machine learning techniques. These approaches attempted to automate pest identification by analyzing features such as color, texture, and shape.

However, they often struggle to perform accurately under real-world agricultural conditions, where factors like varying lighting, complex backgrounds, and overlapping pests make detection difficult. Moreover, these traditional computational methods lack the capability to operate in real-time and are not robust enough to handle diverse pest species, limiting their practical application in modern farming. In recent years, the emergence of deep learning and computer vision technologies has provided new opportunities to overcome these challenges. Advanced models such as YOLO (You Only Look Once) have demonstrated remarkable performance in real-time object detection tasks. Enhancements like attention mechanisms and residual connections further improve detection accuracy and feature extraction, especially for small and complex objects. These technological advancements have paved the way for developing intelligent agricultural systems that not only detect pests efficiently but also provide actionable insights, such as pesticide recommendations. This

project is built upon these innovations to create a smart, automated, and sustainable pest management solution for modern agriculture.

1.1.4 MOTIVATIONS

The primary motivation for this project arises from the increasing challenges faced by farmers due to pest infestations, which significantly reduce crop yield and quality. Traditional pest detection methods rely heavily on manual inspection, which is time-consuming, labor-intensive, and often inaccurate, especially in large agricultural fields. Farmers are frequently unable to detect pests at an early stage, allowing infestations to spread rapidly and cause severe economic losses. This situation highlights the urgent need for an automated and efficient system that can assist in early pest identification and reduce dependency on manual efforts. Another key motivation is the limitation of existing technological solutions in handling real-world agricultural complexities.

Basic image processing and conventional machine learning approaches fail to provide reliable results under varying environmental conditions such as lighting changes, background noise, and overlapping pests. These systems are not suitable for real-time applications and often struggle with detecting small or visually similar insects. Therefore, there is a strong need to adopt advanced deep learning techniques, such as enhanced YOLO models, that can provide high accuracy, robustness, and real-time performance in practical farming scenarios. Additionally, the project is motivated by the need to promote sustainable and eco-friendly agricultural practices. Farmers often rely on generalized pesticide usage due to lack of proper guidance, leading to excessive chemical application, environmental pollution, and increased production costs. By integrating an intelligent pesticide recommendation system with real-time pest detection, the project aims to provide precise and timely guidance for pest control. This not only improves crop protection but also minimizes chemical misuse, supports environmental sustainability, and enhances overall agricultural productivity.

1.2 APPLICATIONS OF THE PROJECT

Smart Agriculture Monitoring: The system can be used for continuous monitoring of crops in smart farming environments, enabling real-time pest detection and helping farmers take timely actions to protect their crops.

Drone-Based Field Surveillance: It can be deployed on agricultural drones to scan large farm areas quickly and efficiently, detecting pest infestations that are difficult to identify manually.

Greenhouse Management: The project can be applied in greenhouse environments to monitor plant health and detect pests early, ensuring controlled and optimized crop growth conditions.

Mobile-Based Pest Detection: Farmers can use mobile devices to capture images of crops and instantly detect pests along with recommended pesticide solutions, making the system accessible and user-friendly.

Precision Farming Systems: The system supports precision agriculture by providing accurate pest identification and targeted pesticide application, reducing resource wastage and improving productivity.

Agricultural Research and Analysis: Researchers can use the system to study pest behavior, infestation patterns, and the effectiveness of pesticides, helping in the development of better pest management strategies.

1.3 CHALLENGES IN THIS PROJECT

- Collecting a large and diverse dataset of pest images with accurate labeling under different environmental conditions.
- Detecting very small, overlapping, or camouflaged pests in complex agricultural backgrounds.

- Handling variations in lighting, weather conditions, and image quality during real-time detection.
- Achieving high accuracy while maintaining real-time performance on devices like drones and mobile phones.
- Avoiding overfitting and ensuring the model generalizes well to unseen pest species and field conditions.
- Integrating an accurate and reliable pesticide recommendation system with proper dosage and safety guidelines.
- Ensuring low computational cost and efficient deployment in resource-constrained environments.
- Managing multiple pest detections simultaneously in a single frame without performance degradation.

1.4 METHODOLOGY

The methodology of the proposed system begins with comprehensive data collection and preprocessing to build a robust dataset for pest detection. Images of various pest species are collected from agricultural fields, drones, public datasets, and research sources under different environmental conditions. These images are then pre-processed through resizing, normalization, and noise removal to improve quality and consistency.

Data augmentation techniques such as rotation, flipping, and brightness adjustments are applied to increase dataset diversity and enhance model generalization. The prepared dataset is carefully labeled and divided into training, validation, and testing sets to ensure effective learning and evaluation. In the next stage, the enhanced YOLO-based deep learning model is developed and trained using the pre-processed dataset.

The model is improved by integrating attention mechanisms and residual connections, which help in capturing important features and improving detection accuracy, especially for small, overlapping, and visually similar pests. During training, hyperparameters such as learning rate, batch size, and epochs are optimized to achieve the best performance.

The model learns to identify pest locations through bounding boxes and classify them into specific categories. Validation techniques are used to monitor the model's performance and prevent overfitting, ensuring reliable results in real-world scenarios. Finally, the trained model is deployed for real-time pest detection and integrated with an intelligent pesticide recommendation system. Input images or live video streams are processed to detect and classify pests instantly

1.4.1 HARDWARE REQUIREMENTS

- Processor : Dual core processor 2.6.0 GHZ
- RAM : 4GB
- Hard disk : 320 GB
- Compact Disk : 650 Mb
- Keyboard : Standard keyboard
- Monitor : 15 inch color monitor

1.4.2 SOFTWARE REQUIREMENTS

- Operating system : Windows OS
- Front End : Html, CSS, JAVASCRIPT
- Back End : Python

- Data base : MySQL SERVER
- IDLE : Python 2.7 IDLE

1.4.4 TECHNOLOGIES USED

Deep Learning

Deep learning is the core technology used in this project for automatic pest detection and classification. It enables the system to learn complex patterns and features directly from raw image data without relying on manual feature extraction. Convolutional Neural Networks (CNNs), a subset of deep learning, play a crucial role in analyzing visual data and identifying pest characteristics such as shape, size, and texture. This allows the system to achieve high accuracy in recognizing different pest species under varying environmental conditions. The strength of deep learning lies in its ability to improve performance with large datasets. As more pest images are used for training, the model becomes more robust and capable of handling diverse real-world scenarios. It can effectively detect pests even when they appear in different orientations, lighting conditions, or backgrounds. This adaptability makes deep learning highly suitable for agricultural applications where environmental variability is high. In this project, deep learning is not only used for detection but also for classification and decision support. The trained model continuously learns and adapts to new data, improving its detection capabilities over time.

YOLO (You Only Look Once)

YOLO is the primary object detection algorithm used in this project due to its speed and efficiency in real-time applications. Unlike traditional detection methods that process images in multiple stages, YOLO performs detection in a single pass, making it extremely fast. This allows the system to process live video

streams and detect pests instantly, which is essential for real-time agricultural monitoring. The enhanced YOLO model used in this project is integrated with attention mechanisms and residual connections to improve detection performance. These enhancements help the model focus on important regions of the image and retain critical feature information, enabling it to detect small, overlapping, and visually similar pests more accurately. This makes YOLO highly effective for complex agricultural environments. Additionally, YOLO provides bounding box predictions along with class probabilities, allowing the system to not only detect pests but also classify them precisely. Its lightweight architecture makes it suitable for deployment on devices such as drones, mobile phones, and edge computing systems, ensuring practical usability in real-world farming scenarios.

1.5 PROBLEM STATEMENT

In the current agricultural scenario, pest infestations continue to pose a serious threat to crop productivity and quality, leading to significant economic losses for farmers. Traditional pest detection methods rely heavily on manual field inspections, which are time-consuming, labor-intensive, and often inaccurate, especially when applied to large-scale farming areas. These methods are not effective in identifying small, overlapping, or camouflaged pests hidden within dense crop environments. As a result, early-stage infestations often go unnoticed, allowing pests to spread rapidly and cause extensive damage before any control measures are implemented. Moreover, existing technological approaches such as basic image processing and conventional machine learning lack the capability to perform accurate and real-time pest detection under varying environmental conditions. Factors such as lighting variations, complex backgrounds, and pest diversity further reduce their effectiveness. In addition, farmers often lack access to timely expert advice for pest control, leading to incorrect pesticide selection, overuse of chemicals, and environmental harm.

Therefore, there is a critical need for an intelligent, automated system that can provide real-time pest detection along with accurate pesticide recommendations to support efficient and sustainable agricultural practices.

1.6 OBJECTIVES

The primary objective of this project is to design and develop an efficient real-time pest detection system using an enhanced YOLO-based deep learning framework for smart agriculture. The system aims to accurately identify and classify various pest species from images and live video feeds captured through drones, mobile devices, or field surveillance cameras. By integrating attention mechanisms and residual connections, the model is designed to improve detection performance, especially for small, overlapping, and visually similar pests in complex agricultural environments. This ensures faster and more reliable monitoring of crop fields compared to traditional manual inspection methods. Another key objective is to support intelligent decision-making in pest management through an automated pesticide recommendation system. Once pests are detected and classified, the system maps them to a predefined knowledge base to suggest appropriate pesticides or fertilizers along with usage guidance. This helps farmers take timely and informed actions, reducing dependency on agricultural experts. Additionally, the project aims to promote sustainable farming practices by minimizing excessive chemical usage, improving crop protection efficiency, and enabling early intervention to prevent large-scale crop damage.

1.7 ABOUT THE PROJECT

This project focuses on developing an intelligent and automated system for real-time insect pest detection and pesticide recommendation in agriculture using an enhanced YOLO-based deep learning framework. The system is designed to

address the limitations of traditional pest monitoring methods by enabling fast and accurate detection of pests directly from images and live video streams. By leveraging advanced deep learning techniques, the system can identify multiple pest species in complex agricultural environments, helping farmers monitor crop health more efficiently and effectively. The core component of the project is an improved YOLO (You Only Look Once) model integrated with attention mechanisms and residual connections. These enhancements allow the model to focus on important regions within an image and retain essential features, improving its ability to detect small, overlapping, and visually similar pests.

The system is capable of real-time operation, making it suitable for deployment on drones, mobile devices, and field surveillance cameras. This ensures continuous monitoring of crops and enables early detection of pest infestations, which is critical for preventing large-scale crop damage. In addition to pest detection, the project incorporates an intelligent pesticide recommendation module that provides suitable treatment suggestions based on the identified pest species. This module uses a knowledge-based approach to recommend appropriate pesticides or fertilizers along with usage guidelines, helping farmers make informed decisions without relying on expert intervention.

By combining detection and decision support, the project promotes sustainable farming practices, reduces excessive chemical usage, and enhances overall agricultural productivity, making it a valuable solution for modern smart agriculture systems.

1.8 RESULT ANALYSIS

The performance of the proposed real-time insect pest detection and pesticide recommendation system is evaluated based on key metrics such as accuracy, precision, recall, F1-score, and processing speed. Compared to

traditional methods and basic image-processing techniques, the enhanced YOLO-based model demonstrates significantly improved detection capability, especially in identifying small, overlapping, and visually similar pests in complex agricultural environments.. Additionally, the system achieves real-time performance, making it suitable for deployment in practical farming scenarios using drones and mobile devices.

Performance Metric	Existing System	Proposed System
Accuracy	70% – 80%	90% – 96%
Precision	65% – 75%	88% – 95%
Recall	60% – 72%	87% – 94%
F1-Score	62% – 73%	88% – 94%
Detection Speed	Slow (Non Real-Time)	Fast (Real-Time)
Small Pest Detection	Poor	High
Overlapping Detection	Low	Strong
Scalability	Limited	High
Chemical Usage Control	Not Available	Intelligent Control

PERFORMANCE COMPARISON

This analysis clearly shows that the proposed system outperforms the existing methods in all major aspects, particularly in accuracy, speed, and intelligent decision support.

1.9 ORGANIZATION OF THE PROJECT

CHAPTER 1: INTRODUCTION

This chapter presents the overall introduction to the project, including the overview, importance, background, and motivation behind developing a real-time

insect pest detection system. It explains the applications, challenges, and methodology adopted in the project. Additionally, it outlines the hardware and software requirements, technologies used, problem statement, objectives, and provides a brief understanding of the project structure and its significance in smart agriculture.

CHAPTER 2: LITERATURE SURVEY

This chapter reviews various existing research works and studies related to pest detection, deep learning, and agricultural automation. It highlights previously proposed methods, techniques, and models used in pest identification and management. The chapter also discusses the limitations of existing approaches, which helps in identifying research gaps and justifies the need for the proposed enhanced YOLO-based system.

CHAPTER 3: SYSTEM STUDY

This chapter explains the analysis of the current system and its drawbacks in pest detection. It clearly describes the existing methods used in agriculture and their disadvantages. Furthermore, it introduces the proposed system and explains how it overcomes the limitations of traditional approaches. The advantages of the proposed system are also discussed in detail.

CHAPTER 4: SYSTEM DESIGN

This chapter focuses on the architectural design of the proposed system. It includes the system architecture diagram and provides a detailed description of each component involved in the process. The chapter explains how data flows through different modules, from input acquisition to pest detection and pesticide recommendation.

CHAPTER 5: SYSTEM IMPLEMENTATION

This chapter describes the practical implementation of the proposed system. It includes the module list and detailed explanation of each module such as data collection, preprocessing, model training, pest detection, and pesticide recommendation. It explains how each module is developed and integrated to form a complete working system.

CHAPTER 6: CONCLUSION AND FUTURE ENHANCEMENT

This chapter summarizes the overall project outcomes and highlights the effectiveness of the proposed system in real-time pest detection and management. It also discusses the potential future enhancements, such as integrating IoT, improving model accuracy, and expanding system capabilities for broader agricultural applications.

APPENDICES

This section includes additional supporting materials such as source code and system screenshots. It provides technical details required for implementation and helps in better understanding the working of the system.

REFERENCES

This section lists all the research papers, journals, and online resources referred to during the development of the project. It ensures proper acknowledgment of existing work and supports the credibility of the proposed system.

CHAPTER 2

LITERATURE SURVEY

2.1 TITLE: InsectMamba: Insect Pest Classification with State Space Model

AUTHOR NAME: Wang, Qianning et al.

YEAR: 2024

DESCRIPTION: This paper proposes a novel insect pest classification approach using a State Space Model (SSM) known as InsectMamba. The study focuses on improving the efficiency and accuracy of pest classification in agricultural environments. Traditional CNN-based models often struggle with long-range dependencies and computational complexity, which limits their performance in real-world applications. To overcome these limitations, the authors introduce a Mamba-based architecture that efficiently captures sequence dependencies with reduced computational cost. The model is designed to handle large-scale insect datasets with high variability in appearance and background conditions. It improves feature representation by modeling global dependencies more effectively than conventional convolutional networks. The approach also enhances classification accuracy for visually similar insect species. Extensive experiments demonstrate that InsectMamba outperforms several existing deep learning models. The model is lightweight and suitable for real-time agricultural applications. It also shows strong generalization across different pest datasets.

The study highlights the importance of state space models in modern pest classification tasks. Overall, it provides a scalable and efficient solution for intelligent pest identification systems.

2.2 TITLE: Different Transfer Learning Approaches for Insect Pest Classification in Cotton

AUTHOR NAME: Toscano-Miranda et al.

YEAR: 2024

DESCRIPTION: This research investigates various transfer learning techniques for insect pest classification specifically in cotton crops. The study evaluates how pre-trained deep learning models can be adapted to agricultural pest datasets with limited training samples. Models such as VGG, ResNet, and Inception are fine-tuned to improve classification performance. The authors emphasize the importance of transfer learning in reducing training time and computational resources. The study addresses challenges such as dataset imbalance and visual similarity among pest species. Data augmentation techniques are used to improve model robustness. The results show that transfer learning significantly enhances classification accuracy compared to training from scratch. The paper also highlights the adaptability of deep learning models to different agricultural conditions.

2.3 TITLE: Machine Learning-Based Approaches for Tomato Pest Classification

AUTHOR NAME: Pattnaik and Parvathy

YEAR: 2022

DESCRIPTION: This paper presents machine learning techniques for classifying pests affecting tomato crops. The study explores traditional classifiers such as Support Vector Machines (SVM), Random Forest, and K-Nearest Neighbors (KNN). Feature extraction methods including texture, color, and shape descriptors are used for pest identification. The research highlights the limitations of manual pest detection in agricultural practices. It focuses on improving classification accuracy using structured feature engineering approaches. The study compares different algorithms to identify the most effective method for pest classification. Results indicate that SVM provides better accuracy compared to other traditional models. However, the performance still depends heavily on

feature quality and preprocessing steps. The paper discusses challenges such as overlapping pests and environmental variations. It concludes that machine learning methods can improve pest identification but lack real-time capability. The study suggests integrating deep learning for better performance in future systems. It contributes foundational knowledge for pest classification research.

2.4 TITLE: A Lightweight Crop Pest Classification Method Based on Improved MobileNet-V2 Model

AUTHOR NAME: Hongxing Peng et al.

YEAR: 2024

DESCRIPTION: This study proposes a lightweight deep learning model based on an improved MobileNet-V2 architecture for crop pest classification. The main goal is to develop an efficient model suitable for mobile and edge devices. The authors optimize the network structure to reduce computational complexity while maintaining high accuracy. Depthwise separable convolutions are used to improve efficiency. The model is trained on diverse pest datasets collected from agricultural environments. Data augmentation techniques are applied to enhance generalization ability. The results show that the improved MobileNet-V2 achieves high classification accuracy with reduced model size. The study emphasizes real-time applicability in smart farming systems. It also demonstrates robustness under varying environmental conditions. The model is suitable for deployment on drones and mobile applications.

2.5 TITLE: Crop Pest Image Classification Based on Improved Densely Connected Convolutional Network

AUTHOR NAME: Hongxing Peng et al.

YEAR: 2023

DESCRIPTION: This paper introduces an improved DenseNet-based convolutional neural network for crop pest image classification. The model focuses on enhancing feature reuse and improving gradient flow through dense connections. The architecture is designed to capture fine-grained pest features for better classification accuracy. The authors modify the traditional DenseNet structure to improve efficiency and reduce redundancy. The model is trained on agricultural pest datasets containing multiple pest categories. Experimental results show improved accuracy compared to standard CNN models. The study highlights the importance of deep feature extraction for distinguishing similar pest species. Data preprocessing and augmentation techniques are used to improve robustness. The model performs well under complex field conditions with varying backgrounds. It demonstrates strong capability in handling multi-class classification problems.

2.6 TITLE: Insect Pest Image Recognition: A Few-Shot Machine Learning Approach Including Maturity Stages Classification

AUTHOR NAME: Gomes, Jacó C. and Díbio L. Borges

YEAR: 2022

DESCRIPTION: This paper proposes a few-shot machine learning approach for insect pest image recognition, focusing on situations where limited labeled data is available. The study also incorporates maturity stage classification of pests, which adds an additional layer of precision for agricultural decision-making. Traditional deep learning models often require large datasets, but this approach reduces dependency on extensive training samples. The model is designed to learn effectively from a small number of annotated images. It improves generalization by leveraging meta-learning techniques. The system is capable of distinguishing pests at different developmental stages, which is crucial for timely intervention. Experimental results show that few-shot learning performs well in

data-scarce agricultural scenarios. The study highlights the importance of adaptive learning strategies in precision agriculture. It demonstrates improved classification efficiency even with limited datasets. The approach is particularly useful for emerging pest outbreaks. It reduces annotation costs and data collection efforts. Overall, it provides a scalable solution for early-stage pest recognition systems.

2.7 TITLE: An Efficient Mobile Model for Insect Image Classification in Field Pest Management

AUTHOR NAME: Zheng, Tengfei et al.

YEAR: 2023

DESCRIPTION: This research focuses on developing an efficient lightweight model for insect image classification suitable for mobile-based field pest management systems. The study aims to provide real-time pest detection capabilities on resource-constrained devices such as smartphones and edge devices. The proposed model reduces computational complexity while maintaining high classification accuracy. It is optimized for deployment in real-world agricultural environments where internet connectivity may be limited. The authors use deep learning optimization techniques to balance speed and performance. The model is trained on diverse insect datasets collected from field conditions. Experimental results show improved inference speed compared to traditional CNN architectures. The system supports real-time decision-making for farmers in pest control activities.

2.8 TITLE: Soybean Images Dataset for Caterpillar and Diabrotica Speciosa Pest Detection and Classification

AUTHOR NAME: Mignoni, Maria Eloisa et al.

YEAR: 2022

DESCRIPTION: This paper introduces a publicly available soybean pest image dataset designed for detecting caterpillar and *Diabrotica speciosa* pests. The dataset includes high-quality images captured under real field conditions, representing different pest appearances and crop environments. It is designed to support research in pest detection and classification using machine learning and deep learning models. The dataset includes annotated images to facilitate supervised learning approaches. The authors emphasize the importance of dataset availability in improving agricultural AI research. The dataset covers variations in lighting, background, and pest positioning. It is suitable for training deep learning models such as CNN and YOLO-based architectures. The study provides baseline experiments to validate dataset usability. Results show that deep learning models perform effectively using this dataset.

2.9 TITLE: Machine Learning for Detection and Prediction of Crop Diseases and Pests: A Comprehensive Survey

AUTHOR NAME: Domingues, Tiago et al.

YEAR: 2022

DESCRIPTION: This paper presents a comprehensive survey of machine learning techniques used for detecting and predicting crop diseases and pests. It analyzes various approaches including traditional machine learning and deep learning models applied in agriculture. The study highlights the evolution of AI techniques in pest detection systems over time. It compares different algorithms based on accuracy, computational efficiency, and scalability. The survey identifies key challenges such as dataset imbalance, environmental variability, and real-time deployment issues. It also discusses the integration of IoT and AI in smart agriculture systems. The paper emphasizes the importance of early detection for minimizing crop losses. It provides insights into future research directions in agricultural AI. The study shows that deep learning models

outperform traditional approaches in most scenarios. It also highlights the need for lightweight models for edge deployment. The survey acts as a foundation for developing advanced pest detection systems. It contributes to understanding current trends in agricultural machine learning research.

2.10 TITLE: MS-ALN: Multiscale Attention Learning Network for Pest Recognition

AUTHOR NAME: Feng, Fuxiang et al.

YEAR: 2022

DESCRIPTION: This paper proposes a Multiscale Attention Learning Network (MS-ALN) for improved pest recognition in agricultural images. The model integrates multiscale feature extraction with attention mechanisms to enhance detection accuracy. It focuses on capturing both global and local pest features for better classification performance. The attention mechanism helps the model focus on important regions containing pests while ignoring background noise. The architecture is designed to handle complex field environments with varying pest sizes.

The model is trained on large pest datasets with diverse categories. Experimental results show significant improvement over traditional CNN-based models. The study highlights the effectiveness of combining multiscale learning with attention mechanisms. It improves recognition of small and overlapping pests. The model demonstrates strong robustness under real-world conditions.

CHAPTER 3

SYSTEM STUDY

3.1 EXISTING SYSTEM

In the current agricultural practices, pest detection is predominantly carried out through manual inspection, where farmers or field workers visually examine crops to identify pest infestations. This approach is highly dependent on human expertise and experience, making it inconsistent and prone to errors. It is also time-consuming and labor-intensive, especially in large-scale farming environments where covering the entire field is difficult. As a result, early-stage infestations are often missed, allowing pests to multiply and spread rapidly before any corrective action is taken. To overcome some of these issues, basic image processing and traditional machine learning techniques have been introduced for pest detection. These methods rely on handcrafted features such as color, shape, and texture to identify pests. However, they suffer from major limitations when applied in real-world agricultural conditions. Variations in lighting, background clutter, weather conditions, and pest appearance significantly affect their accuracy.

Additionally, these systems struggle to detect small-sized pests, overlapping insects, or visually similar species, leading to incomplete or incorrect detection results. Another critical limitation of the existing system is the lack of real-time processing and intelligent decision support. Most traditional and early automated systems are not capable of analyzing data instantly, which delays pest identification and response. Furthermore, there is no integrated mechanism to guide farmers in selecting appropriate pesticides, forcing them to rely on external experts or generalized practices. This often results in incorrect pesticide selection, overuse of chemicals, increased production costs, and environmental damage.

3.1.1 DISADVANTAGES

- Relies heavily on manual inspection, making the process time-consuming and labor-intensive.
- Low detection accuracy due to dependence on human observation and basic image-processing methods.
- Ineffective in detecting small, overlapping, or visually similar pests in complex environments.
- Lacks real-time monitoring capability, causing delays in pest identification and control.
- No integrated pesticide recommendation system, leading to incorrect usage and excessive chemical application.

3.2 PROPOSED SYSTEM

The proposed system presents an advanced and automated solution for real-time insect pest detection using an enhanced YOLO-based deep learning framework. It is designed to overcome the limitations of traditional pest monitoring methods by enabling accurate and fast detection of pests from images and live video streams. The system collects input data from various sources such as drones, mobile devices, and field surveillance cameras, ensuring wide coverage of agricultural fields. By incorporating attention mechanisms and residual connections into the YOLO model, the system improves its ability to detect small, overlapping, and visually similar pests even in complex environmental conditions.

The system operates in real-time, allowing continuous monitoring of crops and immediate identification of pest infestations. The enhanced YOLO model processes each frame efficiently, generating bounding boxes around detected pests and classifying them into specific categories. This real-time capability

ensures that farmers receive instant alerts about pest presence, enabling early intervention and reducing the risk of widespread crop damage. The model is optimized for both accuracy and speed, making it suitable for deployment in resource-constrained devices such as mobile phones and edge-based systems.

In addition to detection, the proposed system integrates an intelligent pesticide recommendation module that provides appropriate treatment suggestions based on the identified pest species. This module uses a knowledge-based mapping system to recommend suitable pesticides or fertilizers along with dosage and safety instructions. By combining detection and decision support, the system reduces reliance on expert consultation and minimizes the chances of incorrect pesticide usage. Overall, the proposed system enhances crop protection, supports sustainable farming practices, and contributes to the development of smart agriculture solutions.

3.2.2 ADVANTAGES

- Enables fast and accurate real-time pest detection using an enhanced YOLO-based model.
- Effectively identifies small, overlapping, and visually similar pests in complex agricultural environments.
- Supports continuous crop monitoring through drones, mobile devices, and surveillance systems.
- Provides intelligent pesticide recommendations with proper guidance, reducing dependency on experts.
- Promotes sustainable farming by minimizing chemical overuse and improving crop protection efficiency.

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

The system architecture of the proposed pest detection framework begins with data acquisition from drones, mobile devices, and field surveillance cameras capturing real-time crop images and videos.

The collected data undergoes preprocessing such as resizing, normalization, and augmentation to improve model performance. The processed data is then fed into the enhanced YOLO-based deep learning model for feature extraction and pest detection. The model identifies and classifies different pest species using attention mechanisms and residual connections for improved accuracy.

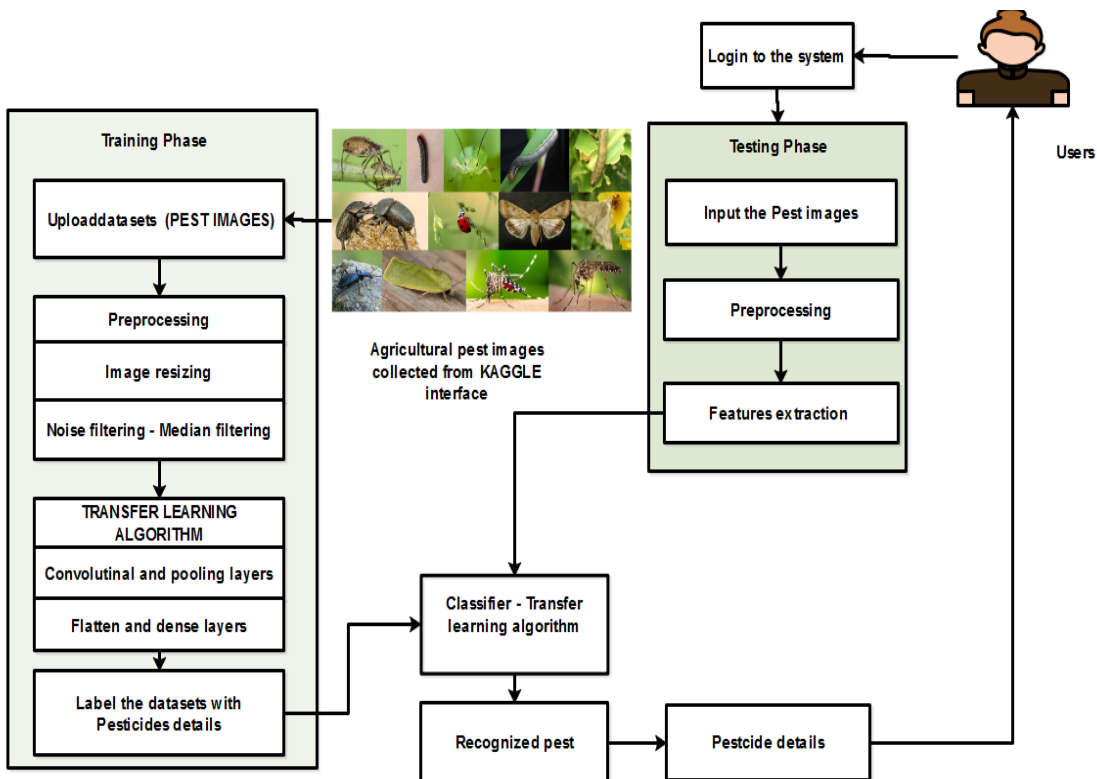


Fig No : 4.1 System Architecture

4.2 ARCHITECTURE DESCRIPTION

The system architecture begins with the data acquisition layer, where images and video streams are captured from various sources such as drones, mobile devices, and field surveillance cameras.

These inputs represent real-time agricultural field conditions and provide continuous monitoring of crops. The collected data is then forwarded to the preprocessing unit, where image enhancement techniques such as resizing, normalization, and noise removal are applied to improve data quality and ensure consistency for further processing.

The processed data is passed to the core detection module, which consists of the enhanced YOLO-based deep learning model. This model is integrated with attention mechanisms and residual connections to improve feature extraction and detection accuracy.

It analyzes the input images to identify pests, draw bounding boxes around them, and classify them into specific categories. The system performs these operations in real-time, enabling quick detection even in complex environments with small or overlapping pests.

After detection, the output is sent to the intelligent pesticide recommendation module. This module maps the identified pest species to a predefined knowledge base and suggests suitable pesticides or fertilizers along with proper usage guidelines.

The final results, including detected pests and recommendations, are displayed to the user through an interface. This complete architecture ensures efficient pest detection, timely decision-making, and supports sustainable agricultural practices.

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 MODULE LIST

- Data Collection
- Pre-Processing
- Model Training with YOLO
- Pest Detection
- Pesticide Recommendation

5.2 MODULE DESCRIPTION

5.2.1 DATA COLLECTION

Data collection is the first and fundamental module of the proposed system, where a comprehensive dataset of pest images is gathered from multiple sources such as agricultural fields, drones, mobile devices, and public research datasets. The collected images include various pest species affecting different crops under diverse environmental conditions. This ensures that the dataset reflects real-world agricultural scenarios.

High-resolution images are preferred to capture fine details of pests. Each image is carefully annotated with proper labels indicating pest categories. The dataset includes both single and multiple pest instances to improve detection capability. Variations in lighting, background, and pest size are considered during collection. This diversity helps the model generalize better. The collected data is organized systematically for easy access and processing. Proper labeling ensures accurate supervised learning. This module lays the foundation for model training. A well-collected dataset directly improves detection accuracy.

5.2.2 PRE-PROCESSING

The preprocessing module prepares the collected images for efficient model training and detection. In this stage, images are resized to a uniform resolution to maintain consistency across the dataset. Noise removal techniques are applied to eliminate distortions and improve image clarity. Normalization is performed to scale pixel values, which helps in faster convergence during training. Data augmentation techniques such as rotation, flipping, zooming, and brightness adjustment are applied to increase dataset size and variability.

This improves the model's robustness to real-world variations. The dataset is split into training, validation, and testing sets. Label encoding is used to convert categorical data into numerical form. Preprocessing ensures that irrelevant features are minimized. It enhances the quality of input data for better feature extraction. This module plays a key role in improving model performance. Proper preprocessing reduces overfitting and increases accuracy.

5.2.3 MODEL TRAINING WITH YOLO

This module focuses on training the enhanced YOLO-based deep learning model using the pre-processed dataset. The YOLO architecture is modified with attention mechanisms and residual connections to improve detection performance. During training, the model learns to identify pest features such as shape, texture, and size. It also learns to draw bounding boxes around detected pests and classify them correctly. Hyperparameters such as learning rate, batch size, and number of epochs are carefully tuned. The training process involves multiple iterations to minimize loss and improve accuracy. Validation data is used to monitor performance and prevent overfitting. Transfer learning techniques may be applied to speed up training. The model learns to handle complex agricultural environments.

5.2.4 PEST DETECTION

The pest detection module is responsible for identifying pests in real-time using the trained YOLO model. Input images or live video streams are captured from devices such as drones, mobile phones, or field cameras. The model processes each frame and detects pests by drawing bounding boxes around them. It classifies each detected pest into a specific category. The system is capable of detecting multiple pests simultaneously in a single frame. It handles small, overlapping, and visually similar pests effectively. Confidence scores are generated for each detection. The results are displayed instantly to the user. This module ensures fast and accurate detection. It supports real-time monitoring of crops. Early detection helps prevent pest spread. This module is critical for timely agricultural intervention.

5.2.5 PESTICIDE RECOMMENDATION

The pesticide recommendation module provides intelligent suggestions based on detected pest species. After detection, the system maps the pest type to a predefined knowledge base. Suitable pesticides or fertilizers are recommended for effective pest control. The system also provides dosage and safety instructions for proper usage. This helps farmers make informed decisions without expert assistance. The module reduces the chances of incorrect pesticide application. It minimizes excessive chemical usage and promotes eco-friendly farming. Recommendations are generated in real-time along with detection results. The system ensures that the selected pesticides are effective for the identified pests. This module improves pest management efficiency. It bridges the gap between detection and action. Ultimately, it enhances crop productivity and sustainability.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

The proposed system for real-time insect pest detection and intelligent pesticide recommendation provides an effective solution to the challenges faced in traditional agricultural practices. By leveraging an enhanced YOLO-based deep learning framework, the system achieves high accuracy and speed in detecting various pest species under complex field conditions. The integration of attention mechanisms and residual connections significantly improves the model's ability to identify small, overlapping, and visually similar pests, making it highly reliable for real-world deployment. The system not only focuses on detection but also enhances decision-making through its intelligent pesticide recommendation module. By automatically suggesting suitable pesticides or fertilizers based on detected pest species, it reduces the dependency on manual inspection and expert consultation. This enables farmers to take timely and appropriate actions, minimizing crop damage and improving overall productivity.

The real-time capability of the system further ensures continuous monitoring and early intervention, which are critical for effective pest management. Overall, the project contributes to the advancement of smart agriculture by combining automation, accuracy, and sustainability. It helps in reducing excessive chemical usage, lowering environmental impact, and promoting eco-friendly farming practices. The system is scalable and can be deployed across different agricultural environments using drones, mobile devices, and surveillance systems. This makes it a practical and impactful solution for modern precision agriculture, supporting better crop protection and improved farmer outcomes.

6.2 FUTURE ENHANCEMENT

The proposed system can be further enhanced by expanding the dataset to include a wider variety of pest species, crop types, and environmental conditions. Incorporating more diverse and high-quality data will improve the model's generalization capability and detection accuracy across different geographical regions. Additionally, advanced deep learning architectures and newer versions of YOLO can be integrated to further enhance performance, especially for ultra-small and highly camouflaged pests. Continuous model updating through real-time data collection can also help the system adapt to evolving pest patterns.

Another potential enhancement is the integration of Internet of Things (IoT) devices and sensor-based monitoring systems. By combining image-based detection with environmental data such as temperature, humidity, and soil conditions, the system can provide more comprehensive pest prediction and prevention strategies. This would enable predictive analytics, allowing farmers to anticipate pest outbreaks before they occur. Furthermore, cloud-based platforms can be used for centralized data storage, processing, and remote monitoring, making the system more scalable and accessible.

The system can also be improved by developing a user-friendly mobile application with multilingual support to ensure accessibility for farmers in different regions. Additional features such as voice assistance, offline functionality, and real-time alerts can enhance usability. Integration with government agricultural databases and expert systems can provide more accurate and region-specific pesticide recommendations.

APPENDICES

A1 SOURCE CODE

```

from flask import Flask, render_template, flash, request, session

import mysql.connector

app = Flask(__name__)

app.config['DEBUG']

app.config['SECRET_KEY'] = '789546321452145a'

@app.route("/")

def home():

return render_template('index.html')

@app.route("/adminlogin")

def adminlogin():

return render_template('Adminlogin.html')

@app.route("/ADMINLOGIN", methods=['GET', 'POST'])

def ADMINLOGIN():

if request.method == 'POST':

if request.form['username'] == 'admin' and request.form['password'] == 'admin':

conn = mysql.connector.connect(user='root', password="", host='localhost',
database='lpestyolodb')

cur = conn.cursor()

cur.execute("SELECT * FROM regtb ")

data = cur.fetchall()

```

```

flash("Your are Logged In...!")

return render_template('AdminHome.html',data=data)

else:

flash("Username or Password is wrong")

return render_template('Adminlogin.html')

@app.route("/userlogin")

def userlogin():

return render_template('Userlogin.html')

@app.route("/USERLOGIN", methods=['GET', 'POST'])

def USERLOGIN():

if request.method == 'POST':

username = request.form['username']

password = request.form['Password']

session['username'] = request.form['username']

conn = mysql.connector.connect(user='root', password="", host='localhost',
database='lpestyolodb')

cursor = conn.cursor()

cursor.execute("SELECT * from regtb where UserName='" + username + "' and
Password='" + password + "'")

data = cursor.fetchone()

if data is None:

flash("Username or Password is wrong...!")

```

```

return render_template('UserLogin.html')

else:

session['email'] = data[3]

conn = mysql.connector.connect(user='root', password="", host='localhost',
database='lpestyolodb')

cur = conn.cursor()

cur.execute("SELECT * FROM regtb where UserName='" + username + "' and
Password='" + password + "'")

data = cur.fetchall()

gtb")

data = cur.fetchall()

return render_template('AdminHome.html', data=data)

@app.route("/predict")

def predict():

import cv2

from ultralytics import YOLO

# Load the YOLOv8 model

flash("Your are Logged In...!")

return render_template('UserHome.html', data=data)

@app.route("/newuser")

def newuser():

return render_template('NewUser.html')

```

```

@app.route("/NEWUSER", methods=['GET', 'POST'])

def NEWUSER():

    if request.method == 'POST':

        name = request.form['name']

        mobile = request.form['mobile']

        email = request.form['email']

        address = request.form['address']

        username = request.form['username']

        password = request.form['password']

        conn = mysql.connector.connect(user='root', password="", host='localhost',
        database='lpestyolodb')

        cursor = conn.cursor()

        cursor.execute("SELECT * from regtb where UserName='" + username + "' and
        Password='" + password + "'")

        data = cursor.fetchone()

        if data is None:

            conn = mysql.connector.connect(user='root', password="", host='localhost',
            database='lpestyolodb')

            cursor = conn.cursor()

            cursor.execute(

                "INSERT INTO regtb VALUES ('" + name + "','" + mobile + "','" + email +
                "','" + address + "','" + username + "','" + password + "')"

```

```

conn.commit()

conn.close()

flash('New User register successfully')

return render_template('Userlogin.html')

else:

flash('Already registered')

return render_template('NewUser.html')

@app.route("/UserHome")

def UserHome():

conn = mysql.connector.connect(user='root', password="", host='localhost',
database='lpestyolodb')

cur = conn.cursor()

cur.execute("SELECT * FROM regtb where
UserName='"+session['username']+"'")

data = cur.fetchall()

return render_template('UserHome.html', data=data)

@app.route("/AdminHome")

def AdminHome():

conn = mysql.connector.connect(user='root', password="", host='localhost',
database='lpestyolodb')

cur = conn.cursor()

cur.execute("SELECT * FROM re

```

```

model = YOLO('runs/detect/pest/weights/best.pt')

# Open the video file

# video_path = "path/to/your/video/file.mp4"

cap = cv2.VideoCapture(0)

ddl = 0

# Loop through the video frames

while cap.isOpened():

    # Read a frame from the video

    success, frame = cap.read()

    if success:

        # Run YOLOv8 inference on the frame

        results = model(frame, conf=0.8)

        for result in results:

            if result.bboxes:

                box = result.bboxes[0]

                class_id = int(box.cls)

                object_name = model.names[class_id]

                print(object_name)

                if object_name != "":

                    ddl += 1

            if ddl == 50:

                ddl = 0

```

```

predicted_class = "

if object_name == "Hellulaundalis":

    out = object_name

    pre = (

        "Control: Spray emamectin benzoate or spinosad. "

        "Remove and destroy infested shoots.\n"

        "Fertilizer: Apply balanced NPK (10:10:10). Avoid excess nitrogen."

    )

    elif object_name == "Leaf Webber":

        out = object_name

        pre = (

            "Control: Use neem oil or chlorantraniliprole. "

            "Install pheromone traps.\n"

            "Fertilizer: Apply vermicompost and potassium-rich fertilizer."

        )

        elif object_name == "ash weevil":

            out = object_name

            pre = (

                "Control: Soil application of chlorpyrifos. "

                "Deep ploughing reduces larvae.\n"

                "Fertilizer: Apply FYM and phosphorus-based fertilizer."

            )

```

```

elif object_name == "blister beetle":

    out = object_name

    pre = (

        "Control: Handpick beetles early morning. Spray carbaryl if severe.\n"

        "Fertilizer: Apply nitrogen in split doses to improve plant recovery."

    )

elif object_name == "fruit fly":

    out = object_name

    pre = (

        "Control: Use methyl eugenol traps. Spray spinosad bait.\n"

        "Fertilizer: Apply potassium-rich fertilizer to improve fruit quality."

    )

elif object_name == "fruit sucking moth":

    out = object_name

    pre = (

        "Control: Use light traps and cover fruits with netting.\n"

        "Fertilizer: Apply balanced NPK and micronutrients."

    )

elif object_name == "helicopter":

    out = object_name

    pre = (

```

```

"Control: Spray emamectin benzoate or indoxacarb.\n"

"Fertilizer: Avoid excess nitrogen. Apply boron and zinc."

)

elifobject_name == "leucinodes":

out = object_name

pre = (

"Control: Remove infested fruits. Spray flubendiamide.\n"

"Fertilizer: Apply phosphorus and potassium fertilizer."

)

elifobject_name == "mealy bug":

out = object_name

pre = (

"Control: Spray imidacloprid or neem oil. Control ants.\n"

"Fertilizer: Apply organic manure and potash."

)

elifobject_name == "pieris":

out = object_name

pre = (

"Control: Spray Bacillus thuringiensis (Bt).\n"

"Fertilizer: Apply nitrogen and calcium fertilizer."

)

elifobject_name == "plutella":

```

```

out = object_name

pre = (

"Control: Use Bt or spinosad. Rotate insecticides.\n"

"Fertilizer: Apply balanced NPK; avoid excess urea."

)

elif object_name == "root grubs":

out = object_name

pre = (

"Control: Soil drenching with chlorpyrifos. Use neem cake.\n"

"Fertilizer: Apply FYM, neem cake, and phosphorus."

)

elif object_name == "schizaphisgraminum":

out = object_name

pre = (

"Control: Spray dimethoate or imidacloprid.\n"

"Fertilizer: Apply potassium-rich fertilizer."

)

elif object_name == "uroleuconcompositae":

out = object_name

pre = (

"Control: Use neem oil or systemic insecticides.\n"

"Fertilizer: Apply NPK with micronutrients."

```

```

)

elif object_name == "whitefly":

    out = object_name

    pre = (

        "Control: Spray acetamiprid or neem oil. Use yellow sticky traps.\n"

        "Fertilizer: Avoid excess nitrogen. Apply potassium and magnesium."

    )

    import winsound

    filename = 'alert.wav'

    winsound.PlaySound(filename, winsound.SND_FILENAME)

    annotated_frame = results[0].plot()

    cv2.imwrite("alert.jpg", annotated_frame)

    sendmail(session['email'], "Pest Name-" + out + "\n Fertilizer" + pre)

    # sendmsg("9486365535", "Prediction Name:" + object_name)

    # Visualize the results on the frame

    annotated_frame = results[0].plot()

    # Display the annotated frame

    cv2.imshow("YOLO11 Inference", annotated_frame)

    # Break the loop if 'q' is pressed

    if cv2.waitKey(1) & 0xFF == ord("q"):

        break

    # Release the video capture object and close the display window

```

```

cap.release()

cv2.destroyAllWindows()

return UserHome()

def sendmsg(targetno,message):

import requests

requests.post(

"http://sms.creativepoint.in/api/push.json?apikey=6555c521622c1&route=transs
ms&sender=FSSMSS&mobilenos=" + targetno + "&text=Dear customer your
msg is " + message + " Sent By FSMSG FSSMSS")

def sendmail(Mailid,message):

import smtplib

from email.mime.multipart import MIMEMultipart

from email.mime.text import MIMEText

from email.mime.base import MIMEBase

from email import encoders

fromaddr = "projectmailm@gmail.com"

toaddr = Mailid

# instance of MIMEMultipart

msg = MIMEMultipart()

# storing the senders email address

msg['From'] = fromaddr

# storing the receivers email address

```

```
msg['To'] = toaddr

# storing the subject

msg['Subject'] = "Alert"

# string to store the body of the mail

# string to store the body of the mail

body = message

# attach the body with the msg instance

msg.attach(MIMEText(body, 'plain'))

# open the file to be sent

filename = "alert.jpg"

attachment = open("alert.jpg", "rb")

# instance of MIMEBase and named as p

p = MIMEBase('application', 'octet-stream')

# To change the payload into encoded form

p.set_payload((attachment).read())

# encode into base64

encoders.encode_base64(p)

p.add_header('Content-Disposition', "attachment; filename= %s" % filename)

# attach the instance 'p' to instance 'msg'

msg.attach(p)

# creates SMTP session

s = smtplib.SMTP('smtp.gmail.com', 587)
```

```
# start TLS for security

s.starttls()

# Authentication

s.login(fromaddr, "kkvzxxkejmebpcyb")

# Converts the Multipart msg into a string

text = msg.as_string()

# sending the mail

s.sendmail(fromaddr, toaddr, text)

# terminating the session

s.quit()

if __name__ == '__main__':

    app.run(debug=True, use_reloader=True)
```

A2 SCREENSHOTS

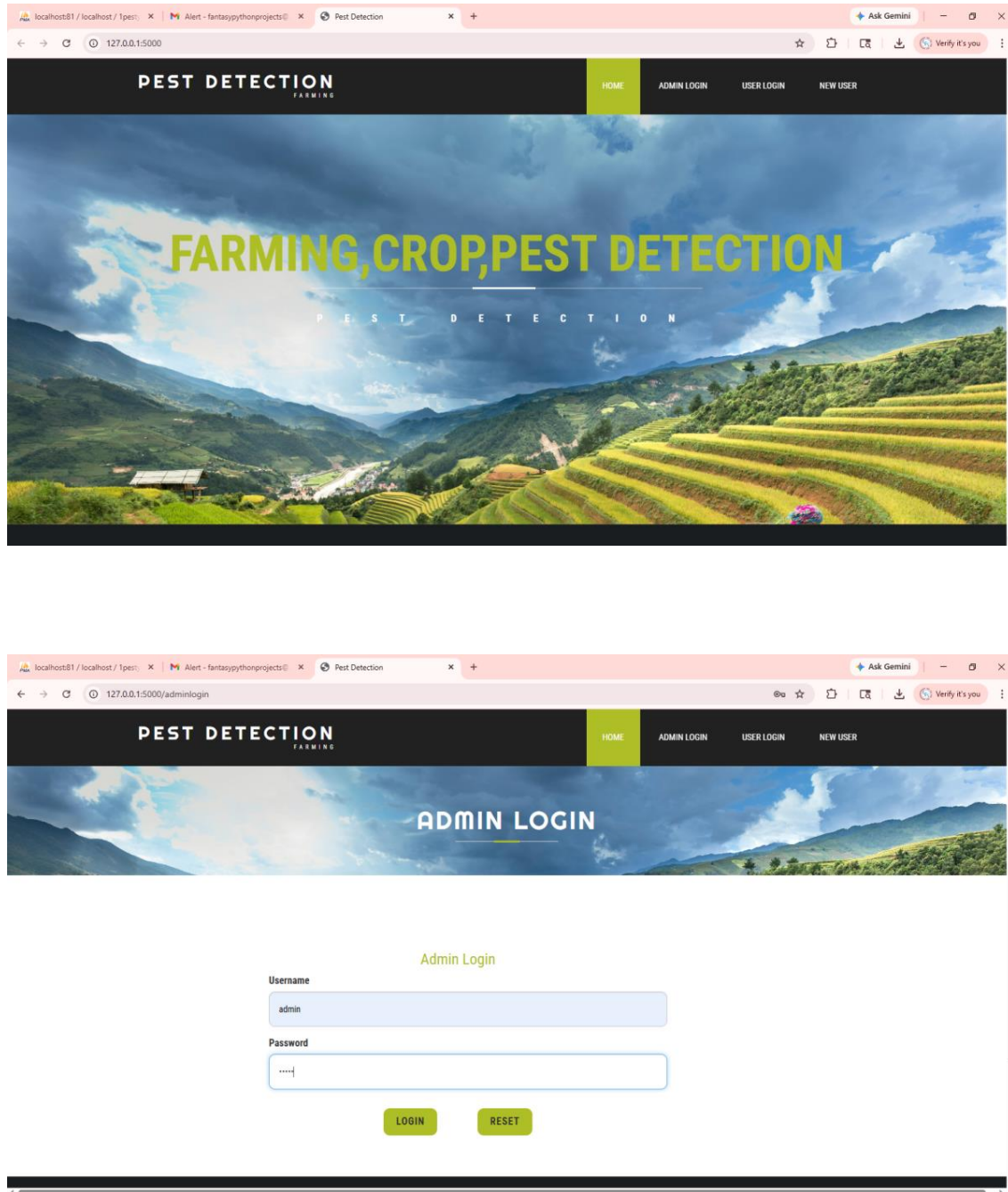


Fig No : A.2 (a) Home Page Screenshot

PEST DETECTION FARMING

ADMIN HOME LOGOUT

USER DETAILS

Name	Mobile	EmailId	Address	UserName
mark	8148526949	fantasypythonprojects@gmail.com	Trichy	mark

© 2026 Pest Detection Farming . All rights reserved | Design by Pest Detection

NEW USER

Name
jack

Mobile
9897885578

Email
fantasypythonprojects@gmail.com

Address
Trichy

Username
jack

Password

SUBMIT RESET

Fig No : A.2 (b) Admin Login Page

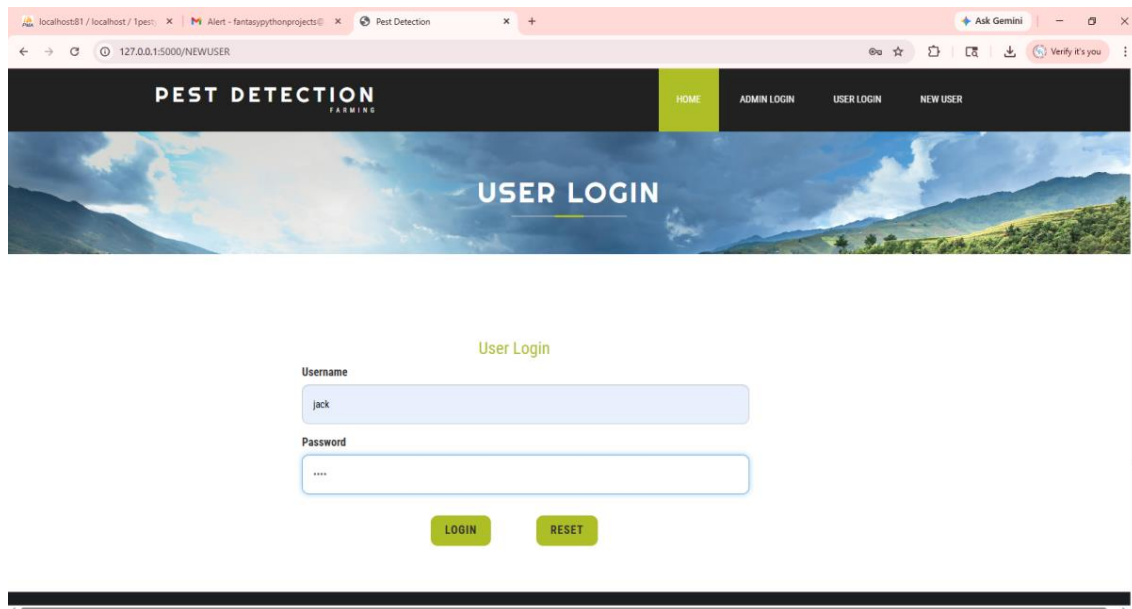
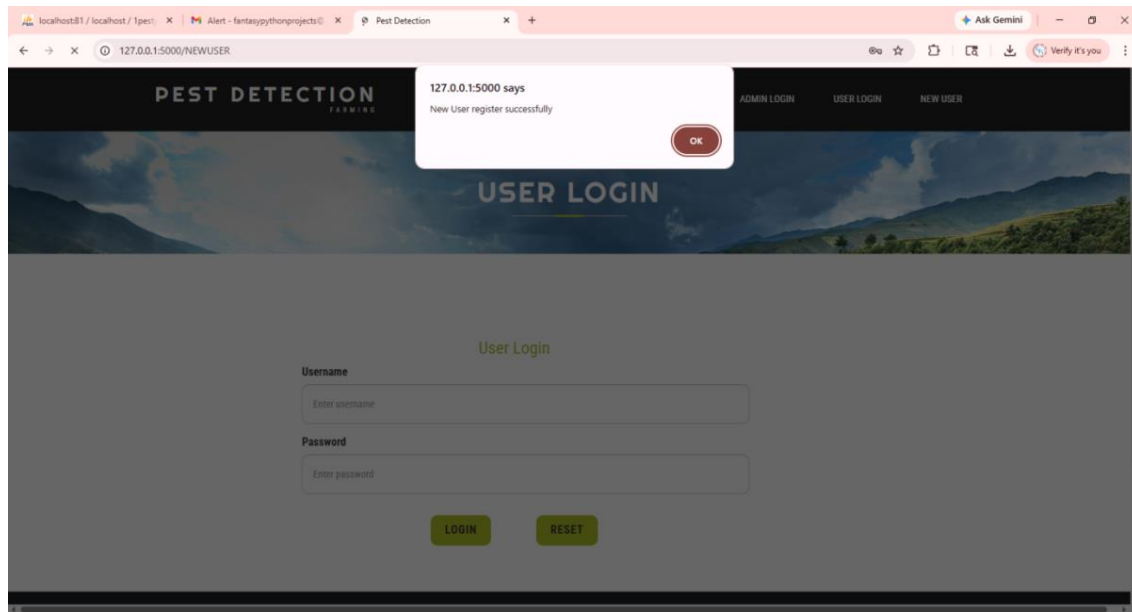


Fig No : A.2 (c) User Registration Page

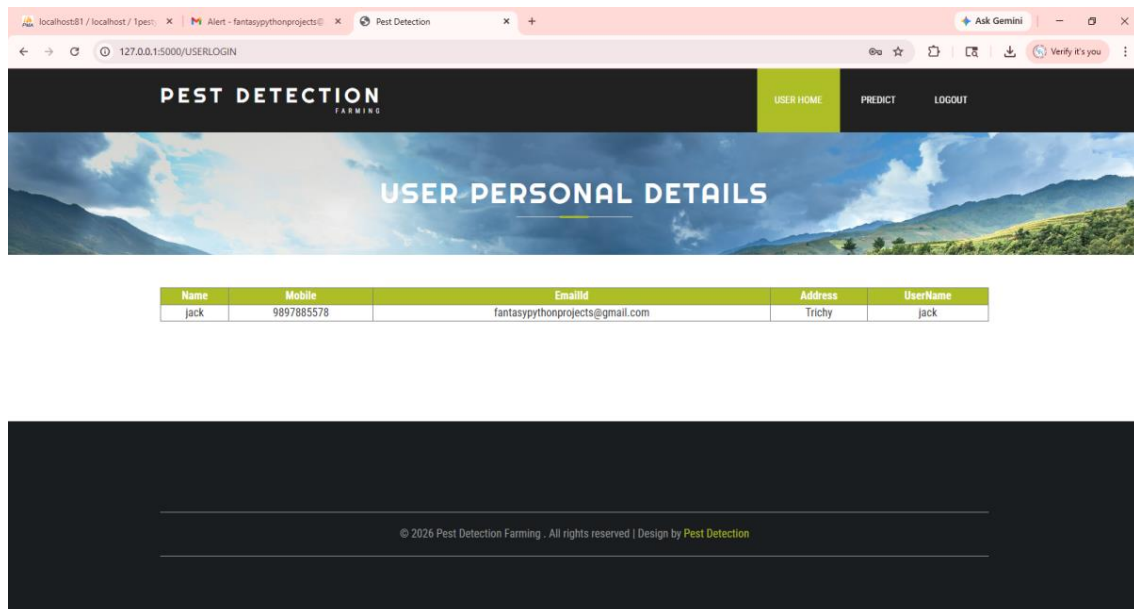
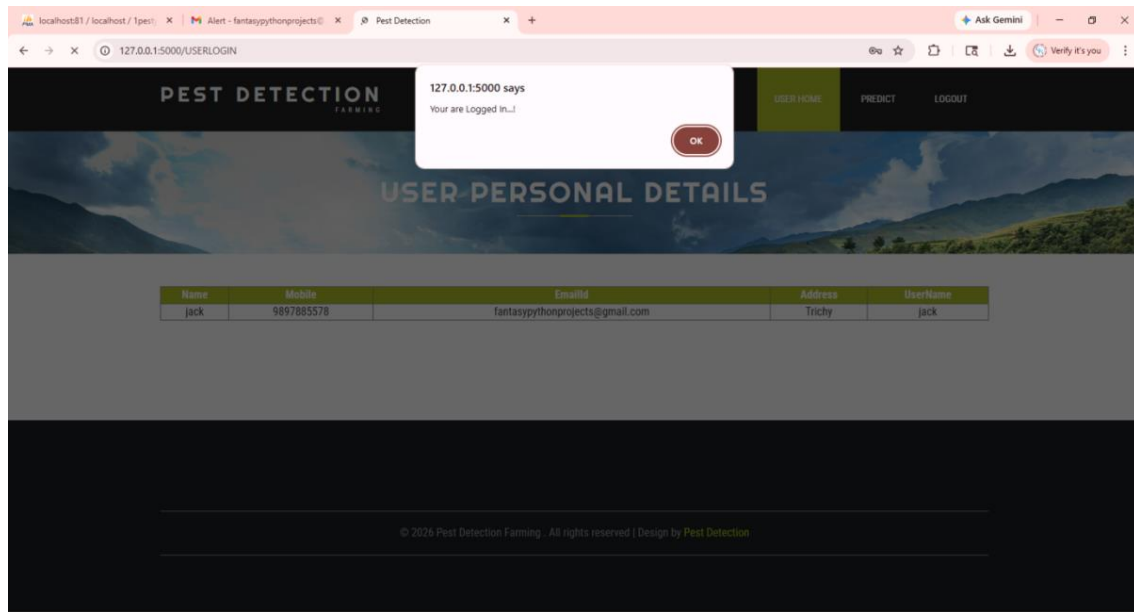


Fig No : A.2 (d) Pest Detection Screen

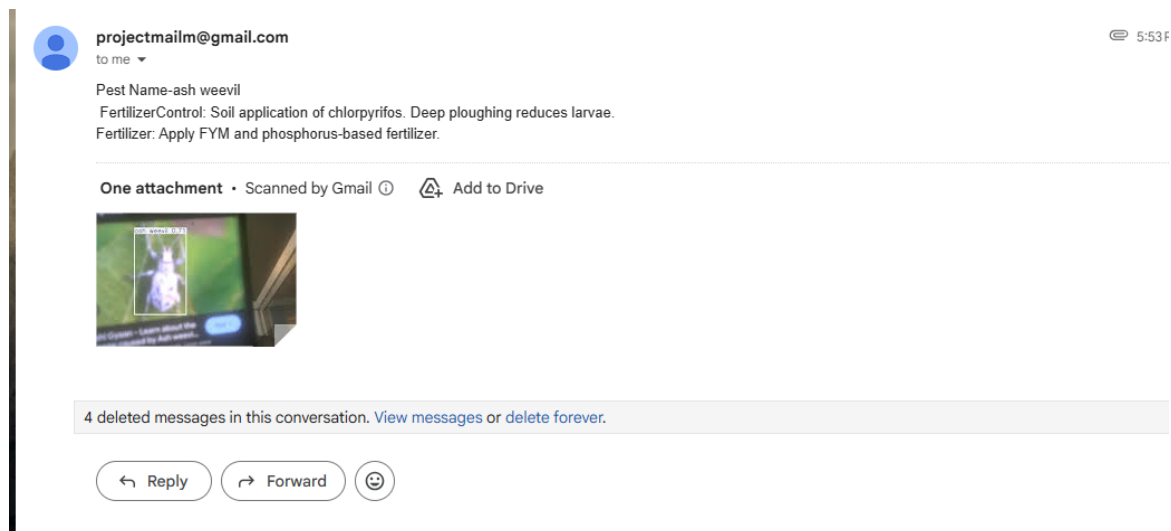
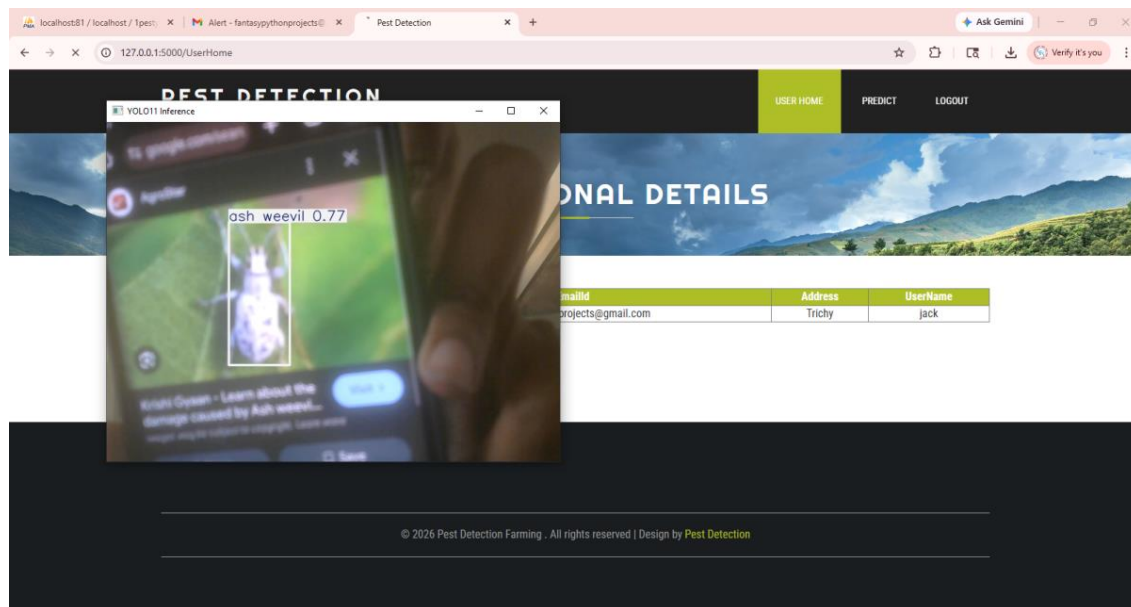


Fig No : A.2 (e) Alert and Recommendation Output

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